Suhas Cristy Mathey

Electrical & Computer Engineer specialized in data science, machine learning, and high-dimensional data analysis. Experienced in scientific computing, signal processing, and developing scalable algorithms for complex sensing, imaging, and embedded data systems.

Email | Website | ms-cristy | Cristy210 | Google Scholar

Education

Master of Science in Electrical & Computer Engineering| *GPA: 3.67/4.0* — Feb 2025
University of Delaware | *Newark, DE, US*
Thesis: *Hyperspectral Image Analysis via Subspace Clustering*
Bachelor of Technology in Electronics & Communication Engineering| *GPA: 7.21/10.0* — May 2021
Indian Institute of Information Technology, Tiruchirappalli | *India*

Technical Skills

Electrical Engineering
- Circuit Analysis
- Embedded Systems
- Sensor Systems
- Hardware Validation

Scientific Computing
- Linear Algebra
- Matrix Decomposition
- Optimization
- Data Visualization

Machine Learning
- NumPy, Pandas
- TensorFlow
- Clustering; PCA
- Neural Networks

Programming
- Python, C++
- Julia, MATLAB
- Anaconda
- Jupyter; Pluto

Tools & Platforms
- Git / GitHub
- Docker
- Linux / Bash
- VS Code

Relevant Coursework

Imaging & Deep Learning, Statistical Learning, Matrix & Tensor Decomposition, Signal Processing, ML & DS
- Explored digital imaging systems, focusing on **mathematical modeling** and **signal processing** (sampling, filtering).
- Topics – Linear/Non-Linear **Regression**, Logistic Regression, Bayesian Analysis; SVMs and **Deep Learning** (ANNs, transformers).
- Hands-on **TensorFlow** labs: training vs. testing, VC dimension, model evaluation.
- Matrix Decomposition – **Eigen Decomposition**, **Singular Value Decomposition**, and their Geometric Interpretation.

Experience

Machine Learning Researcher — Feb 2024 – May 2025
University of Delaware - Electrical & Computer Engineering — Newark, DE, US
- Developed **machine learning algorithms** to analyze high-dimensional sensing datasets (hyperspectral imaging), identifying signal structure, noise characteristics, and latent patterns across large-scale data.
- Implemented scalable data analysis workflows for processing multi-band sensor data, integrating signal processing techniques with statistical analysis.
- **Presented** research findings through weekly lab meetings using data visualizations created using **CairoMakie**, focusing on clear and actionable insights from hyperspectral data analysis.
- Built **reproducible research pipelines** using modular scientific code, version control, and automated testing to support collaborative algorithm development.
- Developed structured experiment protocols and evaluation metrics to test **algorithmic performance** across large sensing datasets.
- Co-developed ***SubspaceClustering.jl***, an open-source software package implementing scalable algorithms for high-dimensional data analysis.

Computational Researcher — Jan 2024 – Feb 2024
University of Delaware - Chemical Engineering — Newark, DE, US
- Implemented algorithms in **Python** for advanced numerical computations and used CuPy to leverage GPU acceleration.
- **Developed** synthetic datasets using **Pandas**, enhancing ML models for accurate prediction of atomic scattering patterns.
- Engineered parallel processing solutions via **Thread Pool Executors** for computational efficiency.

Assistant System Engineer Mar 2022 – Dec 2022
TATA Consultancy Services Chennai, IN

- Dealt with vulnerability assessment checks for the **Citi Corp's** networks and applications.
- Collaborated with a **global** team of 50 in consolidating client's applications security.
- Developed data sheets using advanced **Excel** tools to present data and communicate insights to the **clients**.

Projects

MS Research: Hyperspectral Image Analysis - Unsupervised Learning

Language: Julia, Python, MATLAB

- **Developed and optimized** clustering algorithms in Julia, Python, and MATLAB for pigment mapping, to enhance clustering performance and accuracy in hyperspectral **Reflectance Imaging Spectroscopy (RIS)** datasets.
- Developed and optimized clustering algorithms for analyzing hyperspectral **remote sensing** datasets, including **Pavia, Salinas, Indian Pines, and ONERA**, focusing on **spectral** and **spatial** feature clustering.
- Results from this project have been accepted for a keynote talk at **Techno Heritage 2024** and for poster presentation at **2024 IEEE Baltimore Technical Colloquium**, underlining the contribution to technical innovation in the field of Machine Learning and Signal Processing.

Smart Irrigation system using Internet of Things (IoT) - Capstone Project

Language: C++

- **Designed** and **simulated** an IoT-based irrigation system using Proteus 8 Professional, integrating **hardware sensors** and circuit components with embedded **software control logic**.
- **Programmed** sensors in **Arduino IDE** to collect and analyze soil moisture data, transmitting real-time updates via a **Wi-Fi module** to optimize irrigation based on environmental conditions.
- Performed **data analysis** on sensor feedback to fine-tune water usage, addressing efficiency in water management.
- Developed and integrated both **hardware and software** components into a scalable system architecture, ensuring reliability and energy efficiency for continuous field use.

Feature Engineering with Analytical Data – Dow Chemical (Spring 24)

Language: Python

- Implemented **linear regression models** to analyze spectroscopy datasets of different materials provided by **Dow Chemical**.
- Employed **peak finding** function from the **SciPy library** to accurately detect and pre-process key **spectral features**, improving the precision of data analysis.
- Implemented pre-processing strategies like **normalization** and **dimensionality reduction** to enhance the integrity of the data by minimizing noise and variability amongst the features.
- Leveraged **NumPy** and **SQL** for efficient data manipulation, **querying**, and **analysis** of large spectroscopy datasets, enabling seamless data extraction, pre-processing, and feature selection for **regression analysis**.

Presentations & Talks

Udel GECE Hour – Graduate Student Seminar (Talk) Nov 1, 2024
Mapping Pigments in Reflectance Imaging Spectroscopy (RIS) Datasets via Subspace Clustering

2024 IEEE Baltimore Technical Colloquium Conference (Poster Session & Lightning Talk) Nov 2, 2024
Mapping Pigments in Reflectance Imaging Spectroscopy (RIS) Datasets via Subspace Clustering

Delaware Data Science & Darwin Computing Symposium (Poster Session & Lightning Talk) Feb 13, 2025
Mapping Pigments in Reflectance Imaging Spectroscopy (RIS) Datasets via Subspace Clustering

Awards & Scholarships

2024 Library Student Assistant Scholarship award 2024
Recognized for leadership and excellence in supporting multimedia and digital services – University of Delaware

Graduate Research Assistant Scholarship 2024 – 2025
Awarded full tuition support and stipend for MS research - University of Delaware